



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
COMPUTER ENGINEERING DEPARTMENT
BS COMPUTER ENGINEERING

COURSE INFORMATION SHEET

Session:	Fall 2024
Course Title:	Probability & Statistics
Course Code:	MS-301
Credit Hours:	3
Semester:	5 th
Pre-Requisites:	None
Instructor Name:	Mubin Anwer Siddiqui
Email and Contact Information:	mmobin@ssuet.edu.pk
WhatsApp Group	MS-301Prob&stats
Office Hours:	Monday to Friday (9:00 – 5:00) ✓
Mode of Teaching:	Synchronous/Asynchronous/ Hybrid/Blended

COURSE OBJECTIVE:

This course introduces fundamental concepts in Statistics and Probability to engineering students. Statistics and probability are two important sciences for dealing with variability, uncertainty and complexity in engineering, business, and many real-life applications. This course will provide students with a balanced presentation of introductory mathematical foundations and applications of statistics and probability with emphasis on data analysis and inference. Care has been taken in developing theoretical foundations of statistical and probabilistic techniques with emphasis on simple engineering applications.

COURSE OUTLINE:

Descriptive Statistics:

Basic definitions, measures of central tendency and variation, z-scores, frequency distribution, graphical representation of data, symmetry and skewness, quintiles (percentiles, deciles and quartiles). Linear regression models and correlation coefficient.

Probability theory: Basic definitions and rules of probability, conditional probability, counting techniques.

Random variable: Concept of random variable, discrete and continuous random variable. Discrete and continuous probability distributions: Uniform, binomial, multinomial, negative binomial, geometric, Poisson, Normal; normal approximation and exponential distributions and their applications.

Inferential Statistics:

Estimation of parameters such as mean and variance. Sampling distribution of mean, t-distribution and sampling procedures. z-test, t-test and goodness of fit test.



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COURSE LEARNING OUTCOMES (CLOs) and its mapping with Program Learning Outcomes(PLOs):

CLO No.	Course Learning Outcomes (CLOs)	PLOs	Bloom's Taxonomy
1	Apply descriptive techniques of statistical methods on various data types, regression and correlation.	PLO_1 (Knowledge for solving computing problem)	C3 (Applying)
2	Explain basic concepts and methods of probability theory, random variables, mean and variance, probability distributions (Discrete and Continuous)	PLO_1 (Knowledge for solving computing problem)	C2 (Understanding)
3	Apply technique for sampling, Hypothesis testing, and goodness of fit test.	PLO_2 (Problem Analysis)	C3 (Applying)

RELATIONSHIP BETWEEN ASSESSMENT TOOLS AND CLOs:

Assessment Tools	CLO-1	CLO-2	CLO-3
Quizzes	03 (9%)	04 (9%)	03 (11%)
Assignments	03 (9%)	04 (9%)	03 (11%)
Midterm Exam	20 (63%)	10 (24%)	-----
Final Exam	06 (19%)	24 (58%)	20 (78%)
Total	32	42	26

GRADING POLICY:

Assessment Tools	Percentage
Quizzes	10%
Assignments	10%
Midterm Exam	30%
Final Exam	50%
TOTAL	100%

RECOMENDED BOOKS:



SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY
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- ⑦ Ronald E. Walpole, Sharon L. Myers, and Keying Ye, *Probability & Statistics for Engineers & Scientists*, 9th Edition. Pearson Higher Education.(2017).

ISBN 9789813131279 9813131276

<http://www.elcom-hu.com/Mshtrk/Statistics/9th%20txt%20book.pdf>

- Ross, Sheldon M. “**Introduction to Probability & Statistics For Engineers and Scientist**”, Academic Press, 6th Edition, 2021, ISBN- 978-0-12-824346-6
https://ssuetkhimy.sharepoint.com/:f:/r/personal/fzamir_ssuet_edu_pk/Documents/Books?csf=1&web=1&e=cFWrui

REFERENCE BOOKS:

- Vijay K. Rohatgi, A. K. Md. Ehsanes Saleh, *An Introduction to Probability and Statistics*, 3rd edition. John Wiley & Sons, Inc. (2015)
- F. M. Dekking, C. Kraaikamp, H. P. Lopuhaa, L. E. Meester, *Modern Introduction to Probability and Statistics*, Springer-Verlag London Ltd. (2005)
- Shahid Jamal, *Statistics Problems & Practice for B.Com.*

LECTURE PLAN

Course Title: Probability & Statistics

Course Code: MS-301



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Week No.	Week Dates	Topics	Required Reading	Key Date
1	14-10-2024 to 18-10-2024	Introduction of Statistics Frequency distribution, Graphical Representations	Ross: Chap#2 Ex. 1-25	
2	21-10-2024 to 25-10-2024	Measures of Central Tendency (for both ungrouped and grouped)	Ross: Chap#2 Ex. 1-25	
3	28-10-2024 to 01-11-2024	Measure of dispersion (Variance, S.D , mean deviation, Quartile Deviation, Skewness & Kurtosis	Ross: Chap#2 Ex. 1-25	Assignment # 1
4	04-11-2024 to 08-11-2024	Linear Regression/ Correlation coefficient Scatter diagram, Regression line, Curve Fitting	Ross: Chap#9 Ex. 1-6	Quiz # 1
5	11-11-2024 to 15-11-2024	Introduction of Probability, set theory, Sample space,	Walpole: Chap#2 Ex. 2.1-2.20	
6	18-11-2024 to 22-11-2024	Counting Techniques, permutation, combination etc.	Walpole: Chap#2 Ex. 2.21-2.48	
7	25-11-2024 to 29-11-2024	Events, kinds of events, probability	Walpole: Chp#2 Ex. 2.51-2.70	Assignment # 2
8	Midterm Examination (02-12-2024 to 06-12-2024)			
9	09-12-2024 to 13-12-2024	Conditional probability.	Walpole: Chp#2 Ex. 2.77-2.100	



(SSUET/QR/111)

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10	16-12-2024 to 20-12-2024	Concept of discrete and continuous random variable, distribution and density function	Walpole: Chp#3 Ex. 3.1-3.20	
11	23-12-2024 to 27-12-2024	Joint Density and Distribution Functions	Walpole: Chp#3 Ex. 3.37-3.50	Quiz # 2
12	30-12-2024 to 03-01-2025	Expected Value and variance of function	Walpole: Chp#4 Ex. 4.1-4.20	
13	06-01-2025 to 10-01-2025	Discrete probability Distributions Uniform, Binomial Distribution Multinomial distribution	Walpole: Chp#5 Ex.5.1-5.19	
14	13-01-2025 to 17-01-2025	Poisson distribution, Negative Binomial, Geometric distribution	Walpole: Chp#5 Ex.5.20-5.26	
15	20-01-2025 to 24-01-2025	Continuous Probability Distributions Normal Distribution, Normal Approximation to Binomial, Exponential Distribution Sampling distribution of mean, t-distribution and sampling procedures	Walpole: Chp#6 Ex. 6.1-6.20 6.21-6.30	Assignment #3
16	24-06-2024 to 28-06-2024	. Hypothesis testing: z-test, t-test and goodness of fit test. goodness of fit test.	Notes will be provided	Quiz # 3
Final Examination (28-01-2025 to 08-02-2025)				

Instructor's Name: Mobin Anwer Siddiqui

Dated : 10th Oct 2024

Sign: